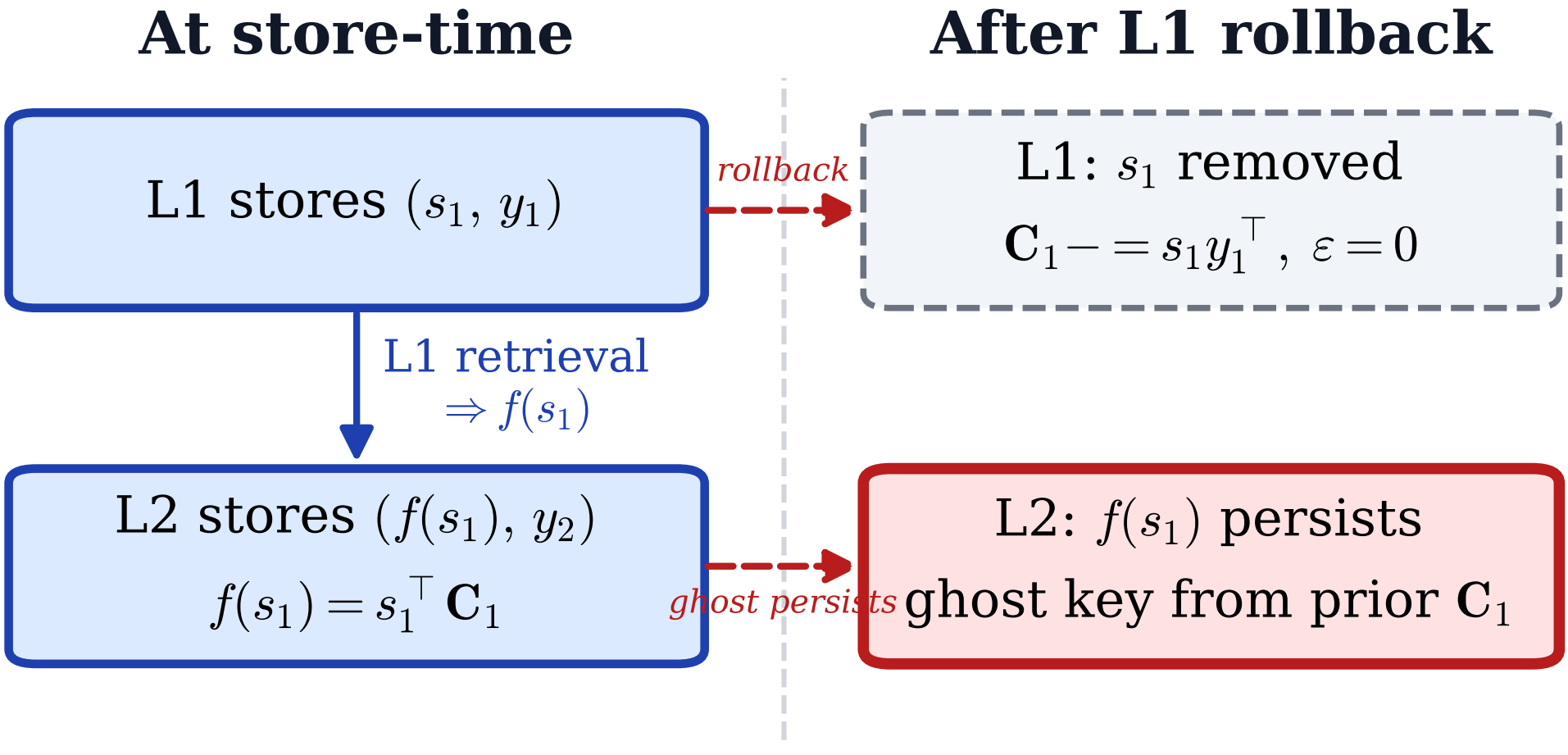
At store-time
After L1 rollback
L1 stores $(s_1, y_1)$
rollback
L1: $s_1$ removed
$\mathbf{C}_1 -= s_1 y_1^\top$, $\varepsilon = 0$
L1 retrieval
$\Rightarrow f(s_1)$
L2 stores $(f(s_1), y_2)$
$f(s_1) = s_1^\top \mathbf{C}_1$
ghost persists
L2: $f(s_1)$ persists
ghost key from prior $\mathbf{C}_1$